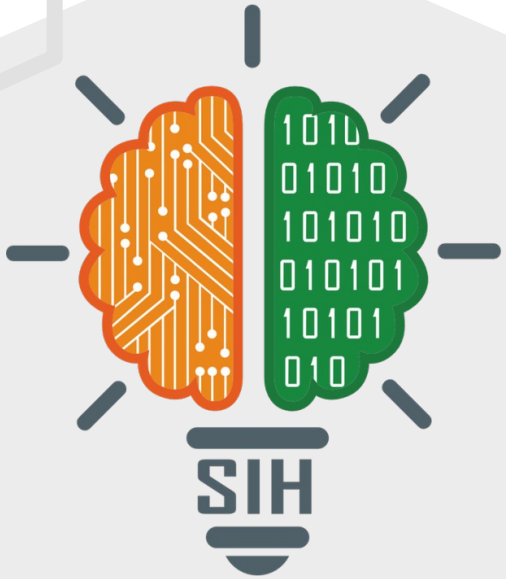
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TITLE PAGE

- **Problem Statement ID:** SIH26249
- **Problem Statement Title:** Air Power — Predictive Maintenance & Fleet Availability (Ministry of Defence)
- **Theme:** *[fill in as listed on the SIH portal]*
- **PS Category:** Software
- **Team Name:** Code Byte
- **Idea: FLIGHTLINE** — Fleet Availability Twin



Code Byte

IDEA TITLE

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FLIGHTLINE — Fleet Availability Twin

A decision layer that joins the four maintenance silos, predicts how many sortie-days the fleet is about to lose, and names the action that saves the most of them — scored in aircraft-days, not RMSE.

THE PROBLEM :

The statement names fragmentation, not accuracy: *"fragmented and largely reactive … not adequately integrated".*

- Health monitoring, technical records, spares and MRO agencies never meet
- A fault predicted on Tuesday collides with a six-week spares lead time — the aircraft sits on the ground anyway
- Cannibalisation is signed without a price
- RUL models report RMSE; nobody reports aircraft-days

THE SOLUTION :

ONE GRAPH OF THE FOUR SILOS
tail → LRU → part → stock → depot bay → sorties

PROGNOSIS WITH HONEST INTERVALS
Remaining life as a 90% conformal interval; abstains when unsure

FLEET SIMULATOR + CP-SAT OPTIMISER
90 days, day by day; re-sequences depot inductions and expedites

RANKED ACTIONS IN AIRCRAFT-DAYS
Each with "Why this number?"; live what-if re-solves in ≈1.0 s

It ranks and prices. It never authorises — the signature stays with the officer.

WHAT IS NEW :

99% vs 32%
of failures warned in time to act (APH), against a capped-RUL model with the better RMSE

+8.9 vs −2.0
cannibalisation priced in aircraft-days: donor already down vs a flying donor

91.6%
interval coverage; when the interval is too wide it says "inspect", not a guess

11.0 pts
of lost availability attributed by cause — 8.0 of them to spares lead time

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TECHNICAL APPROACH

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HEALTH MONITORING
144 installed LRUs · multivariate sensor histories, 6 operating regimes (N-CMAPSS shape)

TECHNICAL RECORDS
14 phase inspections due · durations · cannibalisation history

SPARES
18 stock locations · lead times 7–60 days · orders in transit

MRO AGENCIES
2 depots · 3 bays · one induction crew per depot per day

FLEET TWIN one NetworkX graph · 200 nodes · 392 edges · tail → installed LRU → part number → stock location → depot bay → sortie plan

1 PROGNOSIS
- HistGradientBoosting on rolling mean / std / slope per sensor + age
- Mondrian conformalised quantile regression → 90% RUL interval
- Abstains when interval > decision window

2 SIMULATOR
- 90 days, day by day; wear ∝ sorties flown
- Base-stock (S−1, S) spares, FIFO
- Every down-day = one labelled "hold" → exact attribution

3 OPTIMISER
- OR-Tools CP-SAT: induction day + depot, expedites
- Max sortie-days, then aircraft-days
- 2 passes + simulator check; deterministic budget

4 DECISION LAYER
- Cannibalisation priced over 30 days
- Spare unlock ranking (monotone replay)
- Actions in aircraft-days + causal trace

5 DELIVERY
- FastAPI + offline dashboard
- Live what-if, re-solves in ≈1.0 s
- Printable draft maintenance order

TECHNOLOGY STACK

DATA	ML	OPTIMISATION	GRAPH	SERVING	ASSURANCE
SQLite · seeded generator · N-CMAPSS / C-MAPSS loader	scikit-learn HistGradientBoosting · split-conformal	Google OR-Tools CP-SAT	NetworkX	Python · FastAPI · Tailwind CSS · Chart.js, all bundled	77 pytest tests · same seed → same numbers

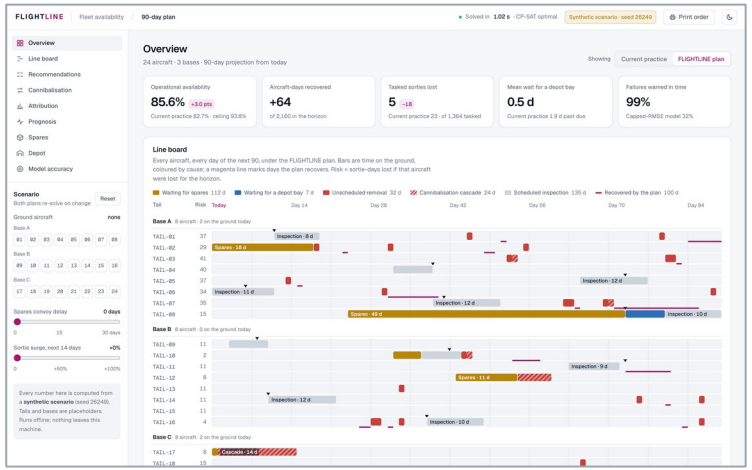
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FEASIBILITY AND VIABILITY

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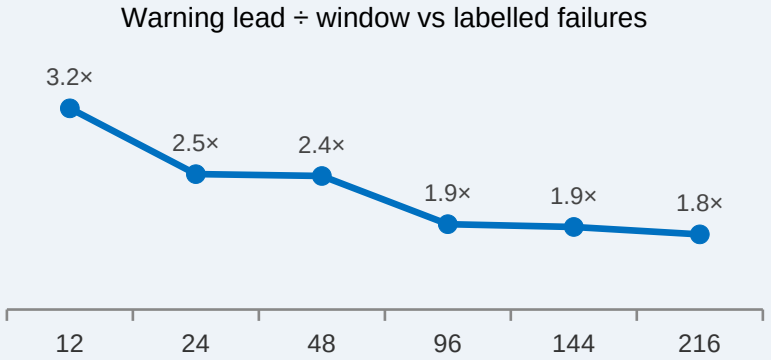
TECHNICALLY FEASIBLE

Built, running and tested — offline



- Runs on one laptop: Python + SQLite; no GPU, no internet
- Both 90-day plans re-simulated and re-solved in ≈1.0 s
- 77 automated tests; same seed → same numbers
- Real N-CMAPSS / C-MAPSS files in data/raw → retrains on real engines
- 2-minute demo film recorded from the running prototype

KEY CHALLENGES	MITIGATION
No IAF data	"SYNTHETIC SCENARIO" badge on every screen; decisions never see ground truth — it is used only to backtest
Model trained on another platform	The decision layer is platform-agnostic: swap the degradation model and the twin, simulator and optimiser are unchanged
Overconfidence near failure	Conformal intervals calibrated per RUL band: 94% coverage in the last 20 cycles; abstains → inspect
Classified data	Air-gapped: one base server, one SQLite file; no cloud, no LLM, no external API
Trust in automated advice	Ranks and prices, never authorises; every number has a trace; every plan re-scored by the full simulator



VIABLE WITH LITTLE DATA

APH stays 95%–100% from 12 to 216 labelled failures. With fewer failures the interval widens and it warns earlier, not later.

More data buys efficiency, not safety.

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IMPACT AND BENEFITS

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+64

aircraft-days in 90 days on a 24-aircraft fleet (Ao 82.7% → 85.6%), same aircraft, parts and bays

23 → 5

tasked sorties lost to unavailability, current practice → plan

1.9 → 0.5

mean days an inspection waits for a depot bay

12 → 8

sorties lost in a 90-day backtest on true failures, capped model → FLIGHTLINE (no prognosis: 18)

Why is availability 82.7% and not 93.7%?

Cause	Ao points	Aircraft-days
Waiting for spares	**8.0**	173
Waiting for a depot bay	**1.3**	27
Unscheduled removal	**1.6**	34
Cannibalisation cascade	**0.1**	3
Gap below the ceiling	**11.0**	**237**

Stress-tested live — each re-solves in about a second

Ground TAIL-01 and TAIL-03
Ao 79.0%; new top action: cannibalise the radar from grounded TAIL-03 to TAIL-02, +28 aircraft-days

Spares convoy 10 days late
Spares share of the gap 8.0 → 9.9 pts; order-now parts 3 → 10

Surge: 40% more sorties
Sorties lost 64 under current practice vs 39 under the plan (+83 aircraft-days)

BASE MAINTENANCE OFFICER

- Why availability is what it is, by cause
- Which depot induction to move, and when

LOGISTICS OFFICER

- Which spare, stocked where, unlocks the most
- Which order to expedite

ENGINEERING OFFICER

- Cannibalisation priced before it is signed
- Inspect or order, with the model's own doubt

AIR-GAPPED DEPLOYMENT

One base server, one SQLite file, open-source stack. Nothing leaves the base network.

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RESEARCH AND REFERENCES

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1 Degradation data

- Arias Chao M. et al. Aircraft engine run-to-failure dataset under real flight conditions (N-CMAPSS). Data 6(1):5, 2021
- Saxena A. et al. Damage propagation modeling for aircraft engine run-to-failure simulation (C-MAPSS). PHM 2008

Used for: Shape of our synthetic degradation; the loader retrains on the real files

2 Conformal prediction

- Romano Y., Patterson E., Candès E. Conformalized quantile regression. NeurIPS 2019
- Vovk V., Gammerman A., Shafer G. Algorithmic Learning in a Random World. Springer, 2005

Used for: 90% remaining-life interval, calibrated per band (Mondrian); the abstention gate

3 Cannibalisation evidence

- US GAO. GAO-02-86, Military Aircraft: Services Need Strategies to Reduce Cannibalizations. Nov 2001
- ≈850,000 cannibalisations and ≈5.3 million extra hours, FY1996–2000; at least twice the labour of a normal repair

Used for: Swap labour priced at 2×; every swap scored before it is signed

4 Spares and optimisation

- Sherbrooke C. METRIC: a multi-echelon technique for recoverable item control. Operations Research 16(1), 1968
- Google OR-Tools CP-SAT · scikit-learn HistGradientBoosting · NetworkX

Used for: Base-stock (S−1, S) spares; induction scheduling and expedites

OUR PILOT AND VALIDATION PLAN

Data
One LRU family: removals, lead times, stock by base, bay counts, cannibalisation logs

→ **Retrain**
Degradation model on real removals; recalibrate the conformal intervals

→ **Shadow mode**
90 days at one base; officers keep deciding, FLIGHTLINE only advises

→ **Measure**
APH and aircraft-days against what actually happened

→ **Expand**
More LRU families, then more bases and types